2

Q2-1

$$h(x) = g \circ f(x)$$

$$f(x) = x^2 + 1, \quad g(x) = \sqrt{x}$$

$$h(x) = g(f(x)) = \sqrt{x^2 + 1}$$

Q2-2

$$f(x) = \sqrt{2x + 1}$$

$$y = \sqrt{2x + 1}$$

$$y^2 = 2x + 1$$

$$x = \frac{y^2 - 1}{2}$$

$$f^{-1}(x) = \frac{x^2 - 1}{2} \quad (x \geq 0)$$

Q2-3

Q2-1

$$h(x) = \sqrt{x^2 + 1} = (x^2 + 1)^{1/2}$$

$$h'(x) = \frac{1}{2}(x^2 + 1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}$$

Q2-4

$$f(x) = \frac{x}{\sqrt{x^2 + 1}}$$

$$f(x) = x(x^2+1)^{-1/2}$$

$$f'(x) = 1 \cdot (x^2+1)^{-1/2} + x \cdot \left(-\frac{1}{2}\right)(x^2+1)^{-3/2} \cdot 2x$$

$$f'(x) = (x^2+1)^{-1/2} - x^2(x^2+1)^{-3/2}$$

$$f'(x) = \frac{x^2+1-x^2}{(x^2+1)^{3/2}} = \frac{1}{(x^2+1)^{3/2}}$$

Q2-5

$$\int \frac{x}{\sqrt{x^2+1}}\, dx$$

Q2-3

$$\sqrt{x^2+1} + C$$